

OPERATING SYSTEMS

ASSIGNMENT - 1

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Write Short Notes on Hardware Protection.

NEED FOR HARDWARE PROTECTION

Early OS were called resident monitors, the OS began to perform many functions especially I/O.

To improve system utilization, OS began to share system resources. Protection is needed for any shared resources.

→ Dual-Mode operation.

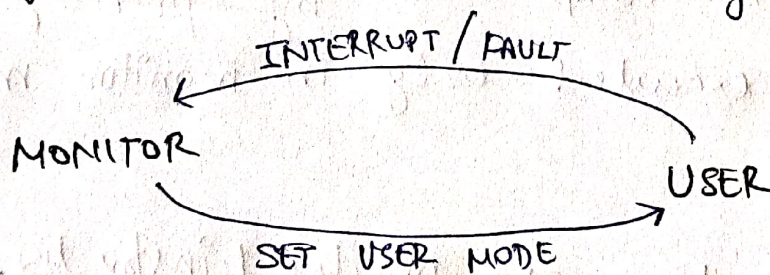
→ I/O protection

→ Memory Protection.

→ CPU Protection

DUAL-MODE OPERATION

* Sharing system resources requires operating system to ensure that an incorrect program can not cause other programs to execute incorrectly.



* User mode - execution done on behalf of user

* Monitor mode (also kernel mode or system mode)
- execution done on behalf of operating system.

* Mode bit added to compute hardware to indicate the current mode: monitor [0] or user [1].

* If bit is set to 1 it is in user mode
If bit set to 0 it is monitor mode.

* Operating system automatically change the bit value according to the program execution.

* When an interrupt or fault occurs hardware switches to monitor mode.

* Initially the hardware starts in monitor mode at system boot time.

* Whenever trap or interrupt occurs, the hardware switches from user mode to monitor mode

MERITS:

* The dual mode operation provides with means of protecting OS from current users.

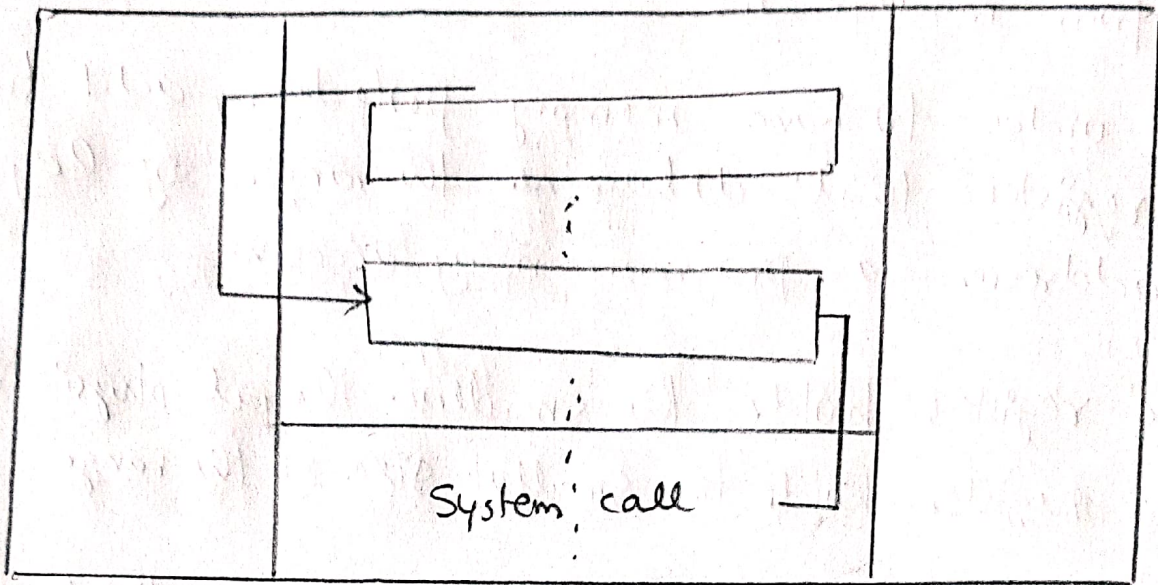
* This protection is accomplished by privileged instructions that can be executed only in monitor mode

I/O PROTECTION

* All I/O instructions are privileged instructions.

* Must ensure that a user program could never gain control of the computer in monitor mode (ie, a user program that, as part of its execution, stores a new address in the interrupt vector.)

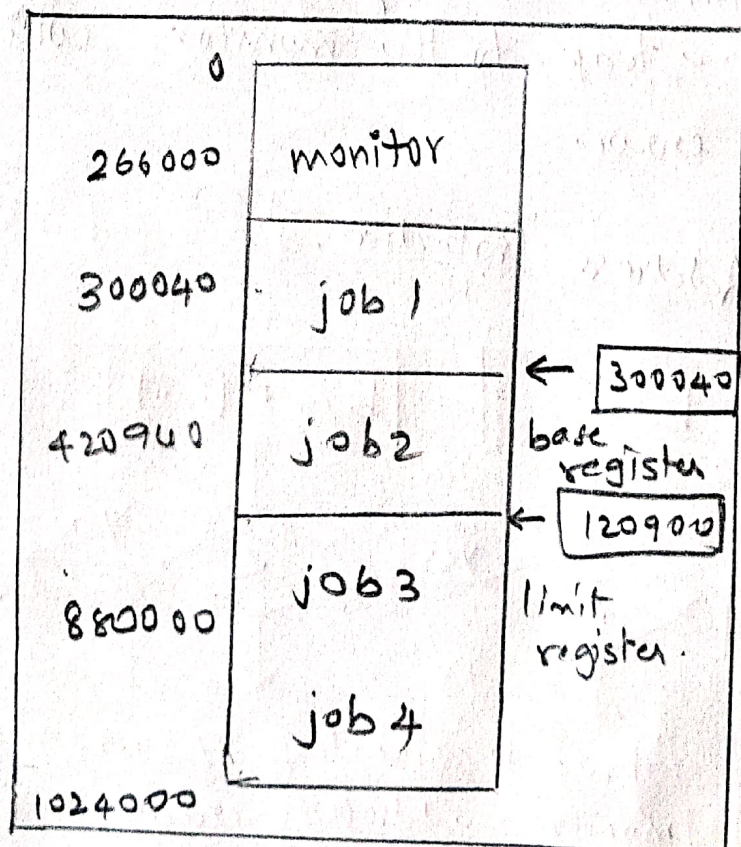
Use of a system call to perform I/O.



Memory protection

Use of a base and Limit Register.

- Must provide memory protection at least for the interrupt vector and the interrupt service routines.



- Also protect interrupt service routines in OS from modification
- In order to have memory protection, add two registers that determine the range of legal addresses a program may access.

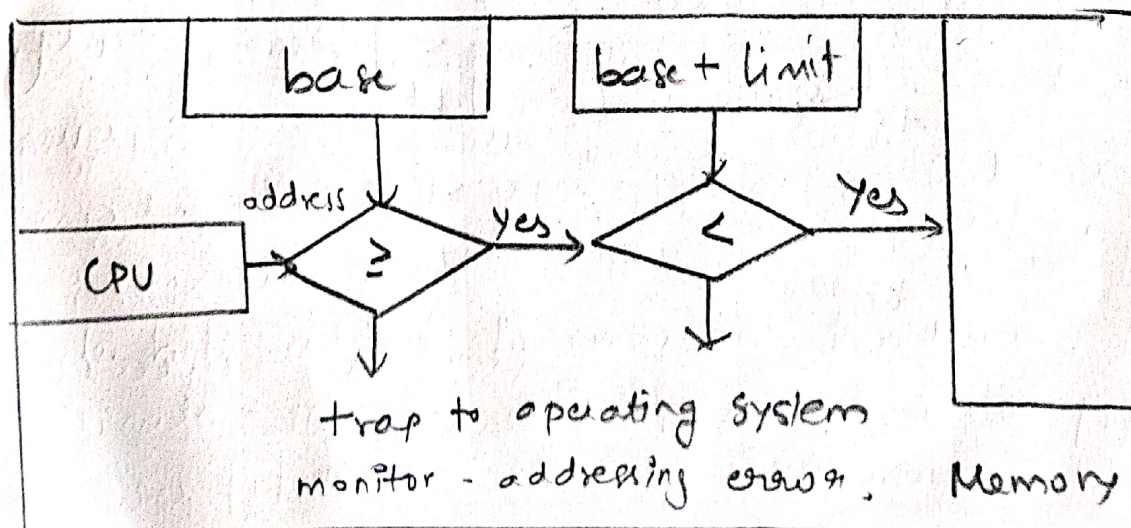
Base register - holds the smallest legal physical memory

Limit register - holds the smallest size of the range.

Eg: If base register holds 300040 and limit range is 120900, then program can legally access all addresses from 300040 through 420940. This schema prevents user program from modifying the code of either OS or other users.

Any attempt by a program executing in users mode to access monitor memory or other users memory results in a trap to the monitor which is a fatal error.

Hardware Address Protection.



HARDWARE PROTECTION

- * When executing in monitor mode, the operating system has unrestricted access to both monitor and user's memory.
- * The load instructions for the base and limit registers are privileged instructions.

CPU PROTECTION

- * Ensure that the OS maintains control
- * Timer can be set to interrupt the computer after a specified period.
- * Period may be fixed or variable.
- * Timer - interrupts computer after specified period to ensure operating system maintains control.
- * Timer is decremented every clock tick.
- * When timer reaches the value 0, an interrupt occurs.
- * Timer commonly used to implement time sharing.
- * Timer also used to compute the current time
- * Load timer is a privileged instruction.